

MEETING AGENDA

Emmy Noether: Symmetry, Conservation, and a Life Against the Barriers

Meeting guide for a talk on Noether's life, context, and physics legacy

Date	Length	Audience	Core theme
29 August 2026	60-75 minutes	General science / cosmology discussion	Symmetry explains conservation

1. Meeting Purpose and Run Sheet

Purpose: Emmy Noether not just as a name attached to a theorem, but as a person who changed physics by showing that conservation laws are the visible face of symmetry.

Session plan

Time	Topic	Meeting objective
0-5 min	Opening hook	Start from a familiar physics question: why should energy stay constant at all?
5-18 min	Early life and Erlangen	Place Noether in family, education, and the exclusion of women from formal standing.
18-32 min	Göttingen and general relativity	Show why Hilbert and Klein needed an invariant-theory expert during the Einstein relativity debate.
32-50 min	Noether's theorem	Explain how time symmetry gives energy conservation, space symmetry gives momentum conservation, and rotational symmetry gives angular momentum.
50-65 min	Nazi dismissal and America	Close the life story: dismissal in 1933, Bryn Mawr, Princeton lectures, and sudden death in 1935.
65-75 min	Discussion	Connect Noether's theorem to current model-building: when a constant changes, what symmetry or conservation law has changed with it?

Opening line to use

The deepest part of physics may not be the equations themselves, but the transformations that leave them unchanged.

- Frame the talk around a human tension: a woman denied normal university status became the person who clarified one of the core structures of modern physics.
- Make the theorem practical: a symmetry is a controlled change in viewpoint that leaves the action unchanged; the conservation law is the bookkeeping that follows.
- Use the recognition story carefully: Noether's 1918 result postdated Einstein's 1915 final GR field-equation paper, but her role in the later energy-conservation problem was long under-centered in popular accounts.

2. Early History: Erlangen, Family, and the Barriers

Historical aim: Show the audience that Noether's mathematical life began inside a gifted but constrained family setting, then developed through informal access, unpaid labor, and intellectual persistence.

Biographical sequence

Date	Event	Talk note
1882	Born Amalie Emmy Noether in Erlangen, Bavaria.	Her father Max Noether was a distinguished mathematics professor at Erlangen.
1889-1900	Educated in girls' schools and certified to teach English and French.	Girls were not given the same preparatory route into university mathematics.
1900-1902	Audited lectures at Erlangen.	Women could attend only informally and with individual professors' permission.
1903-1904	Audited at Göttingen, hearing Hilbert, Klein, Minkowski, Schwarzschild, and others.	This was a first contact with the mathematical culture that later invited her back.
1904-1907	Matriculated at Erlangen after rules changed and completed a doctorate under Paul Gordan.	Her thesis was in classical invariant theory, the technical background that later mattered in relativity.
1908-1915	Worked at Erlangen without pay or formal title.	She advised students and often helped with duties associated with her father, but could not follow the normal male academic path.

Work with Max Noether

- Max Noether's position gave Emmy access to a mathematical household, colleagues, books, and seminar culture, but it did not give her a career track.
- After her doctorate, the habilitation route was effectively closed to her because she was a woman. She stayed in Erlangen, researched, lectured informally, and helped in the institute.
- Students associated with Emmy's teaching and supervision had to be handled through male institutional channels, including her father. This is an important example of invisible labor.

Key people for this section

Person	Role in Noether's life
Max Noether	Father, Erlangen professor, algebraic geometer, and early intellectual environment.
Ida Amalia Kaufmann	Mother, from a Jewish family in Cologne; part of the cultivated family setting.
Paul Gordan	Doctoral adviser, known for constructive invariant theory.
Ernst Otto Fischer	Influenced Noether toward a more abstract algebraic style.

Fundamental Contributions

Period	Role in Noether's life
1907 to 1919	number fields and algebraic invariants.
1920 to 1926	abstract algebra and her theory of mathematical rings
1927 to 1935	noncommutative algebras and hypercomplex numbers

3. Göttingen, Einstein's Theory, and the Recognition Problem

Correction: The university in question is Göttingen, not Göttingen. In 1915 Hilbert and Klein invited Noether there because questions around relativity, invariance, and conservation were becoming urgent.

Why Göttingen mattered

- Göttingen was one of the strongest mathematical centers in the world, shaped by Hilbert and Klein.
- Einstein's general relativity made covariance and invariance central. Conservation of energy and momentum became subtle because gravity was not just a force in space, but part of spacetime geometry.
- Hilbert and Klein saw that Noether's command of invariant theory could clarify the mathematical problem behind general covariance and conservation laws.

Institutional exclusion at Göttingen

Issue	What happened	Talk emphasis
No habilitation at first	Her early attempt to habilitate failed under regulations barring women.	The barrier was not ability; it was formal access denial due to her gender.
Teaching under Hilbert	Courses appeared under Hilbert's name, with Noether assisting.	She did the intellectual work while the institution withheld ordinary standing or attributed to male colleagues
Delayed permission	She became Privatdozent in 1919 and received a modest paid teaching work only later. By no means commensurate to her innate abilities.	Recognition lagged behind contribution.

Einstein, Hilbert, Klein, and Noether

- Einstein's final 1915 general relativity paper did not rest on Noether's theorem, because the theorem came in 1918. Einstein gave physics a dynamical spacetime. But that created a tension in conservation laws (energy). Noether showed that conservation does not come from closed system bookkeeping, rather it comes from symmetry. In GR, the symmetry is deeper than ordinary time-shift symmetry, so the conservation law is also deeper and more subtle.
- The fairer recognition point is that Noether solved the conservation-law difficulty that arose after general relativity was formulated. Her result was mediated through Göttingen's male institutional structures. Felix Klein presented Noether's 1918 paper to the Göttingen society.
- Contemporary and later accounts often gave more attention to Einstein, Hilbert, and Klein with Noether's independent theorem rarely mentioned.

Discussion prompt

**If a physical theory changes the status of time, energy, or constants;
what symmetry principle is being preserved, modified, or abandoned?**

A theory that changes time, energy, or constants must say what transformation still leaves the laws unchanged. That is the Noether question. The conserved quantity follows from the surviving symmetry.

In practical terms:

If the theory changes...	Ask whether this symmetry survives	Consequence
Time or clock-rate	Time-translation symmetry	If preserved, energy conservation remains. If broken, global energy conservation may fail or need redefining.
Space/location	Space-translation symmetry	If preserved, momentum is conserved.
Direction/orientation	Rotational symmetry	If preserved, angular momentum is conserved.
Speed of light / spacetime structure	Lorentz symmetry or general covariance	If modified, the theory must replace standard relativity with a new invariant structure.
Charge or internal fields	Gauge symmetry, e.g. U(1), SU(2), SU(3)	If preserved, charge/current conservation follows.
Mass/inertia/scale	Scale symmetry or a dynamical field symmetry	If masses vary, the theory must explain what remains invariant instead.

Noether teaches us that conservation is not magic bookkeeping. It is the shadow cast by symmetry. If we change time, energy, or constants, we must ask what symmetry still casts that shadow.

For general relativity specifically: ordinary global energy conservation becomes subtle. because spacetime itself changes. Noether's deeper answer is that **diffeomorphism invariance**, the symmetry under smooth coordinate changes, gives local covariant conservation:

$$\nabla_\mu T^{\mu\nu} = 0$$

But a simple global conserved energy exists only when spacetime has a suitable time symmetry, technically a timelike Killing vector.

4. The Theorem: Symmetry as the Source of Conservation

Core explanation: Noether's first theorem stated in 1915; "every continuous differentiable symmetry, of the action of a physical system, has a corresponding conservation law"

→ It defined a fundamental basis of theoretical physics today.

In practical talk language: if the rules do not change under a smooth transformation, something is conserved.

The audience version

Symmetry	Plain meaning	Conserved quantity
Time translation	The laws are the same today, tomorrow, and during the experiment.	Energy
Space translation	The laws are the same here and there.	Momentum
Rotation	The laws are the same after rotating the coordinate frame.	Angular momentum
Gauge symmetry	Certain internal descriptions can change without changing physical observables.	Charge-like conserved quantities and constraints

Solving the time-energy inconsistency in relativity

- In Newtonian thinking, energy conservation can look straightforward because time is an external universal parameter.
- In general relativity, spacetime geometry is dynamical.
A simple global energy conservation statement becomes harder, especially in generally covariant form.
- Noether separated two ideas: finite continuous symmetries produce conservation laws, while infinite-dimensional gauge-type symmetries produce identities among field equations.
- This helped explain why energy conservation in general relativity is not just the old classical theorem in new notation. It is tied to the deeper symmetry structure of the action.

5. Nazi Expulsion, America, Legacy, and Sources

Later Göttingen and the 1933 rupture

- Through the 1920s and early 1930s, Noether became a central figure in modern abstract algebra, influencing students and colleagues far beyond her formal rank.
- After the Nazis came to power in 1933, she was dismissed from Göttingen because she was Jewish. Gender, politics, and antisemitism all compounded earlier barriers.
- Her dismissal was part of the wider destruction of Göttingen's mathematical community and the forced migration of many Jewish and politically vulnerable scholars.

America: Bryn Mawr and Princeton

Date	Event	Meaning
1933	Accepted a visiting professorship at Bryn Mawr College.	For the first time she had a more welcoming academic home.
1934	Gave weekly lectures at the Institute for Advanced Study in Princeton.	Her abstract algebra entered the American research environment.
1935	Died suddenly after surgery in Bryn Mawr, aged 53.	Einstein's public memorial praised her as a creative mathematical genius.

Quotation – Emmy Noether

*My methods are really methods of working and thinking;
this is why they have crept in everywhere anonymously.*

Closing questions...

- What made Noether's theorem so powerful: the algebra, the physics, or the change in what counted as explanation?
- How should the talk handle the injustice without reducing her to the injustice?
- Which bridge matters most for this audience: Symmetry in physics, Women in Universities & Academia, Einstein and relativity, Where is the Noble Prize, Quantum mechanics, or Modern cosmology?

Source notes for the talk

Source	Use in agenda	Link
MacTutor History of Mathematics	Biographical chronology, Erlangen, Göttingen, father, Hilbert/Klein, Bryn Mawr.	Emmy Noether (1882 - 1935) - Biography - MacTutor History of Mathematics
Max Planck Society	Concise overview of career periods, unpaid work, and later expulsion.	"Without Emmy Noether, there would be a huge gap in mathematics and its understanding" Max-Planck-Gesellschaft
Celebratio Mathematica biography	Career barriers, 1920s algebra, Nazi dismissal, Bryn Mawr and IAS context.	Celebratio Mathematica — Noether — Biography
Wikisource: Invariante Variationsprobleme	Original 1918 publication page showing Noether as author and Klein as presenter.	"Without Emmy Noether, there would be a huge gap in mathematics and its understanding" Max-Planck-Gesellschaft

Emmy Noether - NY Times Obituary

by Albert Einstein

The efforts of most human-beings are consumed in the struggle for their daily bread, but most of those who are, either through fortune or some special gift, relieved of this struggle are largely absorbed in further improving their worldly lot. Beneath the effort directed toward the accumulation of worldly goods lies all too frequently the illusion that this is the most substantial and desirable end to be achieved; but there is, fortunately, a minority composed of those who recognize early in their lives that the most beautiful and satisfying experiences open to humankind are not derived from the outside, but are bound up with the development of the individual's own feeling, thinking and acting. The genuine artists, investigators and thinkers have always been persons of this kind. However inconspicuously the life of these individuals runs its course, none the less the fruits of their endeavors are the most valuable contributions which one generation can make to its successors.

Within the past few days a distinguished mathematician, Professor Emmy Noether, formerly connected with the University of Göttingen and for the past two years at Bryn Mawr College, died in her fifty-third year. In the judgment of the most competent living mathematicians, Fräulein Noether was the most significant creative mathematical genius thus far produced since the higher education of women began. In the realm of algebra, in which the most gifted mathematicians have been busy for centuries, she discovered methods which have proved of enormous importance in the development of the present-day younger generation of mathematicians. Pure mathematics is, in its way, the poetry of logical ideas. One seeks the most general ideas of operation which will bring together in simple, logical and unified form the largest possible circle of formal relationships. In this effort toward logical beauty spiritual formulas are discovered necessary for the deeper penetration into the laws of nature.

Born in a Jewish family distinguished for the love of learning, Emmy Noether, who, in spite of the efforts of the great Göttingen mathematician, Hilbert, never reached the academic standing due her in her own country, none the less surrounded herself with a group of students and investigators at Göttingen, who have already become distinguished as teachers and investigators. Her unselfish, significant work over a period of many years was rewarded by the new rulers of Germany with a dismissal, which cost her the means of maintaining her simple life and the opportunity to carry on her mathematical studies. Farsighted friends of science in this country were fortunately able to make such arrangements at Bryn Mawr College and at Princeton that she found in America up to the day of her death not only colleagues who esteemed her friendship but grateful pupils whose enthusiasm made her last years the happiest and perhaps the most fruitful of her entire career.

ALBERT EINSTEIN.

Princeton University, May 1, 1935.

[New York Times May 4, 1935]

Appendix

Meaning... $\nabla_\mu T^{\mu\nu} = 0$

Energy and Momentum are locally conserved in curved spacetime.

Breakdown:

Symbol Meaning

$T^{\mu\nu}$	The stress-energy tensor. It describes energy density, momentum density, pressure, stresses, and energy/momentum flow.
μ, ν	Spacetime indices. Each can refer to time or one of the three space directions.
∇_μ	Covariant derivative. It is the curved-spacetime version of a derivative, adjusted for the geometry of spacetime.
$= 0$	No local creation or destruction of total energy-momentum. Any change is balanced by flow, pressure, stress, or geometry.

$T^{\mu\nu}$ is like the full accounting ledger of matter, radiation, fields, and pressure.

The index ν tells you which conservation equation you are looking at:

ν choice... Rough meaning

$\nu = 0$ Energy conservation equation

$\nu = 1, 2, 3$ Momentum conservation equations in the three spatial directions

The μ index is summed over automatically. So the equation says:

sum over all spacetime directions of the covariant change/flow of energy-momentum = 0

In flat spacetime this is close to: ordinary divergence of energy-momentum = 0

However under general relativity spacetime is curved, so the derivative must include corrections from the metric/connection.

The subtle point:

This is **local conservation**, not necessarily a simple global conserved “total energy of the whole universe.”

Locally, energy-momentum is conserved at every spacetime point. But in an expanding or dynamically curved universe, there may be no universal time symmetry, so there may be no single global energy number that stays fixed forever!

Phd Thesis and Background Publications (German)

[N1]E. Noether: Über die Bildung des Formensystems der ternären biquadratischen Form [On complete systems of invariants for ternary biquadratic forms].

Ph.D. thesis, Friedrich-Alexander-Universität Erlangen-Nürnberg, December 1907. Advised by P. A. Gordan.

[N2]E. Noether: “Rationale Funktionenkörper” [Rational function fields], Jahresber. Deutsch. Math.-Verein. 22 (1913), pp. 316–319. JFM 44.0496.02

[N3]E. Noether: “Körper und Systeme rationaler Funktionen” [Fields and systems of rational functions], Math. Ann. 76 : 2–3 (1915), pp. 161–196. MR 1511817 JFM 46.1442.08

[N4]E. Noether: “Invariante Variationsprobleme” [Invariant variational problems], Nachr. Ges. Wiss. Gött., Math.-Phys. Kl. (1918), pp. 235–257. JFM 46.0770.01

[N5]E. Noether and W. Schmeidler: “Moduln in nichtkommutativen Bereichen, insbesondere aus Differential- und Differenzenausdrücken” [Modules in non-commutative domains, especially those composed of differential and difference expressions], Math. Z. 8 : 1–2 (1920), pp. 1–35. MR 1544424 JFM 47.0097.03

[N6]E. Noether: “Idealtheorie in Ringbereichen” [The theory of ideals in ring domains], Math. Ann. 83 : 1–2 (1921), pp. 24–66. MR 1511996 JFM 48.0121.03